



Finite horizon partially observable semi-Markov decision processes under risk probability criteria

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ARTICLE INFO

Keywords:

Partially observable semi-Markov decision processes
Risk probability criterion
Finite horizon
Optimal Markov policy

ABSTRACT

This paper deals with a risk probability minimization problem for finite horizon partially observable semi-Markov decision processes, which are the fairly most general models for stochastic dynamic systems. In contrast to the expected discounted and average criteria, the optimality investigated in this paper is to minimize the probability that the accumulated rewards do not reach a prescribed profit level at the finite terminal stage. First, the state space is augmented as the joint conditional distribution of the current unobserved state and the remaining profit goal. We introduce a class of policies depending on observable histories and a class of Markov policies including observable process with the joint conditional distribution. Then under mild assumptions, we prove that the value function is the unique solution to the optimality equation for the probability criterion by using iteration techniques. The existence of (ϵ)-optimal Markov policy for this problem is established. Finally, we use a bandit problem with the probability criterion to demonstrate our main results in which an effective algorithm and the corresponding numerical calculation are given for the semi-Markov model. Moreover, for the case of reduction to the discrete-time Markov model, we derive a concise solution.

Partially observable Markov decision processes (POMDPs) are fairly general models to describe stochastic dynamic systems, which are generalizations of traditional Markov decision processes (MDPs). Different from the traditional MDPs with completely observable history information, the system state of POMDPs may not be completely observed. POMDPs are the foundation of learning theory and artificial intelligence, and have wide applications in many fields, including maintenance, signal processing, robotics, and medicine among others. For example, an inventory control with incomplete records can be formulated as a partially observable Markov decision model [8] and the work [6] utilizes a POMDP to determine the optimal maintenance strategy.

There is considerable literature on POMDPs focusing on expected discounted/average criteria [1,8,9,11,12,17,21] and expected exponential utility criteria [2,5,7,10,16,18]. These papers study POMDPs via the technology of augmenting state space. More precisely, they transform the partially observable problem into a completely observable problem by extending the state space with the conditional distribution of the unobservable part of the state, given the observable history. In this way, the POMDPs can be reformulated as standard MDPs with joint states consisting of the observable state and belief state.

Apart from expected discounted/average criteria and exponential utility criteria, risk probability criterion is another worthy investigated optimality criterion in MDPs. The risk probability criterion describes the risk performance of the system and has practical application significance. Actually, in the fields of finance and engineering, probability is often selected as a risk metric, which quantifies the risk probability that the total rewards do not exceed a prescribed profit goal during a finite horizon. For example, the security of insurers can be defined as the probability that total discounted dividends of the insurance company reach a given dividend goal before ruin [19]. However, due to the nonlinearity and non-additivity of probability, the classical dynamic programming theory is not applicable to this kind of problems. The research method above of expectation index cannot be applied to the risk probability criterion.

Up to now, most works focus on the perspective of complete observation for risk probability MDPs. For instance, both [20], [4] investigate the risk probability minimization problem for discrete-time discounted MDPs, they show that the value for the risk probability criterion is the unique solution to an optimality equation, and there exists an optimal policy for the case of bounded rewards and countable state space.

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<https://doi.org/10.1016/j.orl.2024.107187>

Received 14 October 2023; Received in revised form 5 September 2024; Accepted 18 September 2024

Available online 27 September 2024

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Recently, [19] study the optimal dividend payout problem in the framework of discrete-time risk probability MDPs associates with varying discount factors. These results also can be extended to the framework of semi-Markov decision processes (SMDPs). [15] show that this problem can be recast in the theory of SMDPs by enlarging the state space with the remaining profit goal at each stage. More precisely, they introduce a class of policies depending on histories with the additional reward levels, under which the corresponding probability spaces can be constructed. Then, based on the semi-Markov transition law, suitable conditions are given and under which, the authors prove that the value function is a solution to the optimality equation for the probability criterion by using an iteration technique and give the existence of optimal policies.

In these papers with the risk probability criteria, the initial profit goal is given and the remaining profit goals at every stage can be observed directly. Then the general history-dependent policies with the additional reward levels are defined. However, in some cases, the exact state of a system is not completely observed. Such as in artificial intelligence, one of the most challenging problems is that agents only have partial information about the state and events in the environment when they perform actions. To meet the often demanding of theory and practice, it is required to investigate risk-probability MDPs/SMDPs with partially observable state, which is not explored yet.

In this paper, we study risk probability criteria in the framework of partially observable semi-Markov decision processes (POSMDPs), which can also complement the theoretical system for risk probability MDPs/SMDPs. Our objective is to find an optimal policy among history-dependent policies to minimize the risk probability of total rewards that do not exceed a prescribed profit goal over a finite horizon. Due to the challenges caused by the partial observability of the SMDPs, this paper is different from the analysis and optimization approaches of the existing literature of complete observation perspective with risk probability criteria.

We first formulate a fairly general model of controlled POSMDPs with the risk probability criterion which describes the risk performance of the system. Suppose that only one of two components of a controlled semi-Markov process can be observed. However, the reward and the transition law may also depend on both components, which leads to the situation that the remaining profit goal at each stage is an unobservable quantity. Referring to the risk-sensitive problem in [2], we proceed in a similar way for the risk probability criteria by enlarging the state space. We consider the POSMDPs by introducing an information process, and the corresponding problem with complete observation can be reformulated. More precisely, we derive a recursive formula to generate a sequence of probability measures which represents the joint conditional distributions of the unobservable state and the remaining profit goal so far, given the observable history of the process. By enlarging the state space consisting of the observable part of the state and the belief state (the joint conditional distribution above), we establish standard SMDPs. Moreover, based on the belief state, we define Markov policies and prove the existence of the optimal Markov policies.

The rest of the paper is organized as follows. In Section 2, we present the POSMDPs and consider the finite horizon risk probability optimality problem. Moreover, we reformulate a classical semi-Markov decision model by enlarging state space. In Section 3, under continuity and compactness assumptions, we prove that the value function is the unique solution of the optimality equation by using an iteration technique, and we establish the existence of an (ϵ) -optimal policy. Finally, a bandit problem with unobserved states under the probability criterion is considered to illustrate our results. Moreover, an effective algorithm and the corresponding numerical calculation are given.

1. Problem description

The model of POSMDPs consists of the following elements:

$$\{E_X \times E_Y, A, (A(x) : x \in E_X), Q, Q_0, R_0, r\} : \quad (1)$$

- $E_X \times E_Y$ is the Borel state space. For each $(x, y) \in E_X \times E_Y$, the x -component can be observed and the y -component cannot be observed by the decision maker;
- A is the Borel action space with a Borel σ -algebra $\mathcal{A}$; $A(x) \subseteq A$ is the set of actions available for observable state $x \in E_X$. Let $\mathbb{K} := \{(x, y, a) | (x, y) \in E_X \times E_Y, a \in A(x)\}$;
- Q is the semi-Markov kernel on $\mathbb{R}_+ \times E_X \times E_Y$ given $\mathbb{K}$ where $\mathbb{R}_+ := [0, +\infty)$, which describes the transition mechanism of the controlled POSMDPs and satisfies
 - (i) $Q(\cdot, B | x, y, a)$, for any fixed $B \in \mathcal{B}(E_X \times E_Y)$, is a nondecreasing and right-continuous real function on $\mathbb{R}_+$ such that $Q(0, B | x, y, a) = 0$ where $\mathcal{B}(X)$ stands for the σ -algebra on X ;
 - (ii) $Q(t, \cdot | x, y, a)$, for each fixed $t \in \mathbb{R}_+$, is a sub-stochastic kernel on $E_X \times E_Y$ given $\mathbb{K}$;
 - (iii) $Q(\infty, \cdot | x, y, a) := \lim_{t \rightarrow \infty} Q(t, \cdot | x, y, a)$ is a stochastic kernel on $E_X \times E_Y$ given $\mathbb{K}$. If an action $a \in A(x)$ is adopted in the current state pair $(x, y) \in E_X \times E_Y$, then $Q(t, B | x, y, a)$ denotes the joint probability that the sojourn time in state (x, y) is not greater than $t \in \mathbb{R}_+$, and the next state pair belongs to $B \in \mathcal{B}(E_X \times E_Y)$;
- Q_0 is the initial distribution of the unobserved state $y \in E_Y$;
- R_0 is the initial distribution for the decision maker's profit goal $\lambda \in (-\infty, +\infty)$. If the profit goal is λ_0 , then $R_0(d\lambda) = \delta_{\{\lambda_0\}}(d\lambda)$ where $\delta_{\{\lambda_0\}}(d\lambda)$ is the Dirac measure on $\mathcal{B}(\mathbb{R})$ at λ_0 ;
- $r : \mathbb{K} \rightarrow \mathbb{R}_+$ denotes the nonnegative reward rate.

We now describe the evolution of POSMDPs with a probability criterion. Suppose that the system state is $(x_0, y_0) \in E_X \times E_Y$ at the initial decision time $s_0 := 0$ where x_0 is observable and y_0 cannot be observed by the decision maker, and he/she has a profit goal $\lambda_0 \in \mathbb{R} := (-\infty, +\infty)$ and a planning horizon $t_0 \in \mathbb{R}_+$ in mind, that is, within the planning horizon t_0 , he/she tries to earn total rewards more than λ_0 . Then the decision maker chooses an action $a_0 \in A(x_0)$. As a consequence of this action choice, the system remains in (x_0, y_0) until time s_1 , at which point the system state changes to $(x_1, y_1) \in B$ according to the transition law $Q(\theta_1, B | x_0, y_0, a_0)$ where $\theta_1 := s_1 - s_0$. At time s_1 , a reward $r(x_0, y_0, a_0)\theta_1$ is earned, and thus the remaining profit goal is $\lambda_1 := \lambda_0 - r(x_0, y_0, a_0)(\theta_1 \wedge t_0)$ over the remaining planning horizon $t_1 := [t_0 - \theta_1]^+$, where $[y]^+ := \max\{y, 0\}$. Then an action $a_1 \in A(x_1)$ is chosen and the same sequence of events occurs. The process evolves in this way, and so we get an admissible history $\bar{h}_n$ and an observable history h_n of the POSMDPs up to the n -th decision epoch, respectively, i.e., $\bar{h}_n = (\theta_0, t_0, x_0, y_0, \lambda_0, a_0, \dots, \theta_n, t_n, x_n, y_n, \lambda_n)$, and $h_n = (\theta_0, t_0, x_0, a_0, \dots, \theta_n, t_n, x_n)$, where $\theta_0 = 0$, $(x_m, y_m, a_m) \in \mathbb{K}$, $t_0 \in \mathbb{R}_+$, $t_{m+1} := [t_m - \theta_{m+1}]^+$, $\lambda_0 \in \mathbb{R}$, $\lambda_{m+1} := \lambda_m - r(x_m, y_m, a_m)(\theta_{m+1} \wedge t_m)$, for $m = 0, 1, \dots, n-1$, and $(x_n, y_n) \in E_X \times E_Y$. For convenience, we denote by $\bar{H}_n$ and H_n the sets of all admissible histories $\bar{h}_n$ and all observable histories h_n of the system up to the n -th decision time, respectively, and assume $\bar{H}_n$ and H_n are endowed with Borel σ -algebras.

It is natural to consider the policies which only depend on the observed histories. So we define the following deterministic history-dependent policy for the POSMDPs, which represents the action-selection rules for the decision maker.

Definition 1. A deterministic history-dependent policy is a sequence $\pi = \{g_n\}$ of measurable functions $g_n : H_n \rightarrow A$ such that $g_n(h_n) \in A(x_n)$ for each observable history $h_n \in H_n$. The set of all deterministic history-dependent policies is denoted by Π .

For each $(t, x, y, \lambda) \in \mathbb{R}_+ \times E_X \times E_Y \times \mathbb{R}$ and $\pi = (g_0, g_1, \dots) \in \Pi$, by the well-known Ionescu-Tulcea's theorem, there exists a unique probability measure $\mathbb{P}^\pi_{(t, x, y, \lambda)}$. The measure $\mathbb{P}^\pi_{(t, x, y, \lambda)}$ on $\bar{H}_0$ is given

by $\mathbb{P}^\pi_{(t,x,y,\lambda)}(0, t, x, y, \lambda) = 1$. Suppose $\mathbb{P}^\pi_{(t,x,y,\lambda)}$ has a projection onto $\overline{H}_n$ which has been constructed. Then $\mathbb{P}^\pi_{(t,x,y,\lambda)}$ on $\overline{H}_{n+1}$ is determined as follows:

$$\mathbb{P}^\pi_{(t,x,y,\lambda)}(\Gamma \times (da, ds, \tilde{t}, dx, dy, d\tilde{\lambda})) := \int_{\Gamma} \mathbb{P}^\pi_{(t,x,y,\lambda)}(d\tilde{h}_n) \mathbb{I}_{\{\tilde{t}=[t_n-s]^+\}} \times \delta_{\{\lambda_n-r(x_n, y_n, a)(s \wedge t_n)\}}(d\tilde{\lambda}) Q(ds, dx, dy | x_n, y_n, a) \delta_{\{g_n(h_n)\}}(da).$$

Its corresponding expectation operator is denoted by $\mathbb{E}^\pi_{(t,x,y,\lambda)}$.

Then we can denote the corresponding stochastic processes $\{\Theta_n, S_n, X_n, Y_n, \Lambda_n, A_n, n \geq 0\}$ on H_∞ where $S_n := \sum_{k=1}^n \Theta_k$. Corresponding to the stochastic process $\{S_n, X_n, Y_n, A_n\}_{n \geq 0}$, we can define an underlying continuous time state-action process $\{X(t), Y(t), A(t), t \in \mathbb{R}_+\}$ by $X(t) = \sum_{n=0}^\infty X_n \mathbb{I}_{\{S_n \leq t < S_{n+1}\}} + X_\infty \mathbb{I}_{\{t \geq S_\infty\}}$, $Y(t) = \sum_{n=0}^\infty Y_n \mathbb{I}_{\{S_n \leq t < S_{n+1}\}} + Y_\infty \mathbb{I}_{\{t \geq S_\infty\}}$, and $A(t) = \sum_{n=0}^\infty A_n \mathbb{I}_{\{S_n \leq t < S_{n+1}\}} + A_\infty \mathbb{I}_{\{t \geq S_\infty\}}$, where $X_\infty \notin E_X$, $Y_\infty \notin E_Y$ and $A_\infty \notin A$ denote some isolated points, $\mathbb{I}_E$ is the indicator function on a set E , and $S_\infty := \lim_{n \rightarrow \infty} S_n$.

Now we consider the following optimality problem.

For each (observable) initial planning horizon-state pair $(t, x) \in \mathbb{R}_+ \times E_X$, the initial distribution Q_0 and R_0 of the unobservable state and the profit goal, respectively, we define the risk probability criterion of POSMDPs under a policy $\pi \in \Pi$ by

$$F^\pi(t, x) := \int_{E_Y \times \mathbb{R}} \mathbb{P}^\pi_{(t,x,y,\lambda)} \left(\int_0^t r(X(s), Y(s), A(s)) ds \leq \lambda \right) \times Q_0(dy) R_0(d\lambda),$$

which measures the risk of the system that the total rewards obtained during the planning horizon $[0, t]$ do not exceed some profit goal, when the initial observable state x and the joint conditional initial distribution $Q_0 \otimes R_0$ of the unobservable state and the profit goal are given. So $F^\pi(t, x)$ can be regarded as a risk probability.

Definition 2. A policy $\pi^* \in \Pi$ is said to be optimal if $F^{\pi^*}(t, x) = F^*(t, x)$ for each $(t, x) \in \mathbb{R}_+ \times E_X$, where $F^*(t, x) := \inf_{\pi \in \Pi} F^\pi(t, x)$ is the value function.

The aim of this paper is to give conditions for the existence of an optimal policy.

2. Preliminaries

Note that only one component of every state pair can be observed. However, the reward depends on both components, which leads to the situation that the accumulated rewards incurred so far is an unobservable quantity and so the remaining profit goal at each stage is also an unobservable quantity.

Let $\mathcal{P}(X)$ be the family of probability measures on $\mathcal{B}(X)$. Then we can introduce the following probability measures:

$$\mathbb{P}^\pi_{(t,x,\mu)}(\cdot) := \int_{E_Y \times \mathbb{R}} \mathbb{P}^\pi_{(t,x,y,\lambda)}(\cdot) \mu(dy, d\lambda),$$

where $\mu \in \mathcal{P}(E_Y \times \mathbb{R})$. Its corresponding expectation operator is denoted by $\mathbb{E}^\pi_{(t,x,\mu)}$. The probability measure μ plays the role of the conditional distribution on the larger state space including hidden state component and the remaining profit goal.

Motivated by the new definitions of probability measures above, we redefine the risk probability criterion: for each $(t, x) \in \mathbb{R}_+ \times E_X$ and $\mu \in \mathcal{P}(E_Y \times \mathbb{R})$:

$$V^*(t, x, \mu) := \inf_{\pi \in \Pi} V^\pi(t, x, \mu) \quad (2)$$

$$:= \inf_{\pi \in \Pi} \int_{E_Y \times \mathbb{R}} \mathbb{P}^\pi_{(t,x,y,\lambda)} \left(\int_0^t r(X(s), Y(s), A(s)) ds \leq \lambda \right) \mu(dy, d\lambda)$$

It is obvious that $F^\pi(t, x) = V^\pi(t, x, Q_0 \otimes R_0)$.

To solve the optimality problem, we will construct an updating procedure which can generate a filtered process. We next discuss it in two cases:

(i). For denumerable state space, we define the updating operator:

$$\Psi(t, x, \mu, a, \theta, t', x')(y', B) := \frac{\phi^*(t, x, \mu, a, \theta, t', x')(y', B)}{\sum_{y \in E_Y} Q(\theta, x', E_Y | x, y, a) \mu^Y(y)},$$

where $\phi^*(t, x, \mu, a, \theta, t', x')(y', B) := \sum_{y \in E_Y} \int_{\mathbb{R}} \int_B Q(\theta, x', y' | x, y, a) \times \delta_{\{\lambda - r(x, y, a)(\theta \wedge t)\}}(d\lambda') \mu(y, d\lambda) \delta_{\{[t-\theta]^+\}}(t') \text{ and } B \in \mathcal{B}(\mathbb{R})$.

(ii). For the uncountable state space, before constructing the updating operator, we first assume that the semi-Markov kernel Q has a measurable density q with respect to some σ -finite measures ν and η , i.e., $\forall t \in \mathbb{R}_+, B \in \mathcal{B}(E_X \times E_Y)$,

$$Q(t, B | x, y, a) = \int_B q(t, x', y' | x, y, a) \nu(dx') \eta(dy').$$

For any $\mu' \in \mathcal{P}(E_Y)$, we define the marginal transition kernel density $q^X(t, x' | x, y, a) := \int_{E_Y} q(t, x', y' | x, y, a) \eta(dy')$. Then, for the uncountable state space, we can define the following updating operator $\Psi : \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R}) \times A \times \mathbb{R}_+ \times \mathbb{R}_+ \times E_X \rightarrow \mathcal{P}(E_Y \times \mathbb{R})$:

$$\Psi(t, x, \mu, a, \theta, t', x')(B) := \frac{\phi^*(t, x, \mu, a, \theta, t', x')(B)}{\int_{E_Y} q^X(\theta, x' | x, y, a) \mu^Y(dy)},$$

where $\phi^*(t, x, \mu, a, \theta, t', x')(B) := \int_{E_Y} \int_{\mathbb{R}} \int_B q(\theta, x', y' | x, y, a) \times \eta(dy') \delta_{\{\lambda - r(x, y, a)(\theta \wedge t)\}}(d\lambda') \mu(dy, d\lambda) \delta_{\{[t-\theta]^+\}}(t') \text{ with } B \in \mathcal{B}(E_Y \times \mathbb{R})$ and $\mu^Y(dy) := \mu(dy, \mathbb{R})$ is the Y -marginal distribution of μ .

Moreover, for any given $\mu \in \mathcal{P}(E_Y)$, if the state space is denumerable, we define $\forall x' \in E_X$, $Q^X(t, x' | x, \mu', a) := \sum_{y \in E_Y} Q(t, x', E_Y | x, y, a) \mu'(y)$. For the uncountable case, $\forall D \in \mathcal{B}(E_X)$, $Q^X(t, D | x, \mu', a) := \int_D \int_{E_Y} q^X(t, x' | x, y, a) \mu'(dy) \nu(dx')$.

Remark 3. (1). $\Psi(t, x, \mu, a, \theta, t', x')(\cdot)$ is a probability measure when $t' = [t - \theta]^+$, that is, $\Psi(t, x, \mu, a, \theta, t', x')(E_Y \times \mathbb{R}) = 1$.

(2). We define the updating operator only when the denominator is positive.

For each $n \geq 0$, $h_n = (\theta_0, t_0, x_0, a_0, \dots, \theta_n, t_n, x_n)$, $B \in \mathcal{B}(E_Y \times \mathbb{R})$ ($B \in E_Y \times \mathcal{B}(\mathbb{R})$ for denumerable state space), we define a sequence of probability measures:

$$\mu_0(B | h_0) := (Q_0 \otimes R_0)(B),$$

$$\mu_{n+1}(B | h_n, a, \theta, t, x') := \Psi(t_n, x_n, \mu_n(\cdot | h_n), a, \theta, t, x')(B). \quad (3)$$

This recursion is called a filtered equation.

Following a similar approach to [1], we can prove that $\{\mu_n\}$ represents a sequence of conditional distributions on the larger state spaces of hidden state component and the remaining profit goal, that is, $\forall n \in \mathbb{N}_0 := \{0, 1, 2, \dots\}$, $\pi = (g_0, g_1, \dots) \in \Pi$, $B \in \mathcal{B}(E_Y \times \mathbb{R})$

$$\mu_n(B | \Theta_0, T_0, X_0, A_0, \dots, \Theta_n, T_n, X_n)$$

$$= \mathbb{P}^\pi_{(t,x)}((Y_n, \Lambda_n) \in B | \Theta_0, T_0, X_0, A_0, \dots, \Theta_n, T_n, X_n). \quad (4)$$

Now we turn again to the optimality problem (2). We also define a suitable model for SMDPs to solve problem (2) for POSMDPs.

For this purpose we consider SMDPs with the state space $E := E_X \times \mathcal{P}(E_Y \times \mathbb{R})$ and the action space A . $A(x, \mu) := A(x)$ denotes the set of all actions available when the system is in state $(x, \mu) \in E$ and $\mathbb{K} := \{(x, \mu, a) | x \in E_X, \mu \in \mathcal{P}(E_Y \times \mathbb{R}), a \in A(x, \mu)\}$ is the set of all measurable subsets of $E \times A$. For each $(x, \mu) \in E$, $a \in A(x, \mu)$, $t \in \mathbb{R}_+$ and a Borel subset $B \subseteq E$, we define the transition law $\tilde{Q}$ as follows: $\forall s \in \mathbb{R}_+$, $\tilde{Q}(t, B | x, \mu, a) := \int_{E_X} \mathbb{I}_B((x', \Psi(s, x, \mu, a, t, [s - t]^+, x')) Q^X(t, dx' | x, \mu^Y, a)$. Note that $\tilde{Q}$ is a semi-Markov kernel. In the SMDPs setting, we define the following deterministic Markov policy.

Definition 4. Let $\mathbb{F}$ denote the set of all measurable functions $f : \mathbb{R}_+ \times E \rightarrow A$ such that $f(t, x, \mu) \in A(x, \mu)$. A deterministic Markov policy is a sequence $\{f_n\}$ of measurable functions $f_n \in \mathbb{F}$ for all $n \in \mathbb{N}_0$. We denote by Π_M the family of all deterministic Markov policies.

Remark 5. Note that “Markov” means that the decision only depends on the current planning horizon $t \in \mathbb{R}_+$ and state $(x, \mu) \in E$. We also have $\Pi_M \subset \Pi$ in the following sense: For each $\pi = (f_0, f_1, \dots) \in \Pi_M$, we can find a policy $\pi' = (g_0, g_1, \dots) \in \Pi$ such that $g_0(\theta_0, t_0, x_0) := f_0(t_0, x_0, \mu_0)$ and

$$g_n(h_n) := f_n(t_n, x_n, \mu_n(\cdot|h_n)) \quad \forall n \in \mathbb{N}.$$

So V^π is also well defined for $\pi \in \Pi_M$.

To solve the finite horizon problem, without loss of generality, we let horizon length be arbitrarily fixed with $T \in \mathbb{R}_+$. In applications, it is natural to avoid the possibility of an infinite number of jumps during a finite horizon. So we develop the following assumption throughout this paper.

Assumption 6. There exist positive constants δ, ϵ_0 such that

$$Q(\delta, E_X, E_Y | x, y, a) \leq 1 - \epsilon_0 \quad \forall (x, y, a) \in \mathbb{K}.$$

Next, we give a statement whose proof can be similarly completed following the arguments of Proposition 2.1 in [14].

Proposition 7. Under Assumption 6, it holds that $\mathbb{P}_{(t,x,y,\lambda)}^\pi(\{S_\infty > T\}) = 1 \quad \forall (t, x, y, \lambda) \in [0, T] \times E_X \times E_Y \times \mathbb{R}, \pi \in \Pi$.

To show the properties of $V^\pi(t, x, \mu)$, we note that for each $(t, x) \in \mathbb{R}_+ \times E_X$, $\mu \in \mathcal{P}(E_Y \times \mathbb{R})$ and $\pi \in \Pi$,

$$\begin{aligned} V^\pi(t, x, \mu) &= \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}_{(t,x,y,\lambda)}^\pi \left(\sum_{m=0}^{\infty} \int_{S_m \wedge t}^{S_{m+1} \wedge t} r(X(s), Y(s), A(s)) ds \leq \lambda \right) \\ &\quad \times \mathbb{I}_{[0,+\infty)}(\lambda) \mu(dy, d\lambda) \\ &= \lim_{n \rightarrow \infty} \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}_{(t,x,y,\lambda)}^\pi \left(\sum_{m=0}^n \int_{S_m \wedge t}^{S_{m+1} \wedge t} r(X(s), Y(s), A(s)) ds \leq \lambda \right) \\ &\quad \times \mathbb{I}_{[0,+\infty)}(\lambda) \mu(dy, d\lambda), \end{aligned}$$

where the first equality follows from Assumption 6, the second equality is due to the nonnegativity of r , the continuity of probability measures, and the dominated convergence theorem.

Moreover, we define

$$\begin{aligned} V_{-1}^\pi(t, x, \mu) &:= \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0,\infty)}(\lambda) \mu(dy, d\lambda) = \int_{\mathbb{R}} \mathbb{I}_{[0,+\infty)}(\lambda) \mu^\Lambda(d\lambda), \\ V_n^\pi(t, x, \mu) &:= \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}_{(t,x,y,\lambda)}^\pi \left(\sum_{m=0}^n \int_{S_m \wedge t}^{S_{m+1} \wedge t} r(X(s), Y(s), A(s)) ds \leq \lambda \right) \\ &\quad \times \mathbb{I}_{[0,+\infty)}(\lambda) \mu(dy, d\lambda) \quad \forall n \geq 0. \end{aligned}$$

Clearly, $V_n^\pi(t, x, \mu) \geq V_{n+1}^\pi(t, x, \mu) \forall (t, x, \mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$, $\pi \in \Pi$, $n \geq -1$. Furthermore, $\lim_{n \rightarrow \infty} V_n^\pi(t, x, \mu) = V^\pi(t, x, \mu)$.

3. Main results

Below we state our main results, we first introduce the following notations. Let $\mathcal{H}$ be the set of Borel-measurable functions $H : \mathbb{R}_+ \times E \rightarrow [0, 1]$ where $E = E_X \times \mathcal{P}(E_Y \times \mathbb{R})$. In addition, for any $H \in \mathcal{H}$, $(x, \mu) \in E$, $a \in A(x)$, $f \in \mathbb{F}$ and $t \in \mathbb{R}_+$, we define operators T^a , T^f and T as follows:

$$\begin{aligned} T^a H(t, x, \mu) &:= \int_{E_Y \times \mathbb{R}} (1 - Q^X(t, E_X | x, y, a)) \mathbb{I}_{[0,\lambda]}(r(x, y, a)t) \\ &\quad \times \mathbb{I}_{[0,\infty)}(\lambda) \mu(dy, d\lambda) \\ &\quad + \int_{E_X} \int_0^t H(t-u, x', \Psi(t, x, \mu, a, u, t-u, x')) \\ &\quad \times Q^X(du, dx' | x, \mu^Y, a), \end{aligned}$$

$$T^f H(t, x, \mu) := T^{f(t,x,\mu)} H(t, x, \mu),$$

$$TH(t, x, \mu) := \inf_{a \in A(x)} T^a H(t, x, \mu).$$

Moreover, for any given $h_n \in H_n$ and $\{g_n\} \in \Pi$, we define $T^{g_n} H(t, x, \mu) := T^{g_n(h_n)} H(t, x, \mu) \quad \forall n \geq 0$. Next we characterize the probability $V^\pi(t, x, \mu)$ by approximating.

Lemma 8. Let $\pi = \{f_0, f_1, \dots\} \in \Pi_M$ be arbitrary. Under Assumption 6, for each $n \geq -1$, we have

(a). $V_n^\pi(\cdot, \cdot, \cdot) \in \mathcal{H}$ and $V^\pi(\cdot, \cdot, \cdot) \in \mathcal{H}$;

(b). $V_{n+1}^\pi(t, x, \mu) = T^{f_0} V_n^{(1)\pi}(t, x, \mu)$, $V^\pi(t, x, \mu) = T^{f_0} V^{(1)\pi}(t, x, \mu)$, where $^{(1)\pi} = \{f_1, f_2, \dots\} \in \Pi_M$.

In particular, $V^f(t, x, \mu) = T^f V^f(t, x, \mu)$ for any $f \in \mathbb{F}$.

Note that T^a and T^f are operators from $\mathcal{H}$ to $\mathcal{H}$, but T may not be. To ensure T to be an operator from $\mathcal{H}$ to $\mathcal{H}$, we propose the following condition.

Assumption 9. (a). $A(x)$ is compact;

(b). $Q^X(t, E_X | x, y, a)$ is lower semicontinuous in $a \in A(x)$;

(c). $\int_{E_X} \int_0^t H(t-u, x', \Psi(t, x, \mu, a, u, t-u, x')) Q^X(du, dx' | x, \mu^Y, a)$ is lower semicontinuous in $a \in A(x)$ for any fixed $H \in \mathcal{H}$.

Theorem 10. Under Assumptions 6 and 9, $\forall (t, x, \mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$, let $F_{-1}^*(t, x, \mu) := \int_{\mathbb{R}} \mathbb{I}_{[0,\infty)}(\lambda) \mu^\Lambda(d\lambda)$, and for every $n \geq 0$, $F_{n+1}^*(t, x, \mu) := T F_n^*(t, x, \mu)$. Then we have

(a). $\lim_{n \rightarrow \infty} F_n^* = V^*$ and $V^* \in \mathcal{H}$.
(b). The value function $V^*(t, x, \mu)$ is the unique solution to the optimality equation: $V^*(t, x, \mu) = TV^*(t, x, \mu)$.
(c). There exists an $f^* \in \mathbb{F}$ such that $V^*(t, x, \mu) = T^{f^*} V^*(t, x, \mu)$ and the policy $\pi^* = f^*$ is optimal.

Definition 11. For any given $\epsilon > 0$, a policy $\pi_\epsilon^* \in \Pi$ is said to be an ϵ -optimal policy if $\|V^* - V^{\pi_\epsilon^*}\| \leq \epsilon$, where the norm $\|\cdot\|$ is defined by $\|h\| = \sup_{(t,x,\mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})} |h(t, x, \mu)| \quad \forall h \in \mathcal{H}$.

Theorem 12. Under Assumptions 6 and 9, for every $(t, x, \mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$, the following statements hold.

(a) For any sufficiently small $\rho > 0$, let $n_0 := [\log_\beta(\rho(1-\beta))]^+ + \bar{k}$, where $[x]^+$ denotes the largest integer not bigger than $x \in \mathbb{R}_+$, $\beta := (1 - \epsilon_0^{\frac{1}{k}})^{\frac{1}{k}}$ and $\bar{k}$ is any fixed nonnegative integer such that $\bar{k} > \frac{T}{\delta}$. Then we have $\|V^* - F_{n_0}^*\| \leq \rho$, $\forall n \geq n_0$.
(b) There exists $f_\epsilon^* \in \Pi$ satisfying $F_{n_0+1}^* = T^{f_\epsilon^*} F_{n_0}^*$ and the policy $\pi_\epsilon^* := (f_0^*, f_1^*, \dots) \in \Pi$ is an $\epsilon := \rho(2-\beta)$ -optimal policy.

Proof. The proof can be similarly completed through the argument of Theorem 4.2 in [13].

4. Bandit problem

We apply the above POSMDP model under the probability criteria to a so-called bandit problem. Suppose that there are two slot machines with unknown success probabilities θ_1 and θ_2 , which follow a given initial distribution. Whenever arm 1 is chosen, the decision maker waits for time $t_1 \in \mathbb{N}_0$ to observe success (is represented by “ $z = 1$ ”) or failure (“ $z = 0$ ”). If arm 2 is chosen, the wait time is t_2 . At each decision time, the decision maker has to choose an arm, and if that arm leads to a win, he/she gets one unit reward, otherwise there is no gain. In addition, suppose that after the last decision, no matter how much time is left, the result can be observed and the corresponding reward is obtained. Now there is a target reward λ_0 , and the decision maker aims to minimize the probability that the total rewards obtained in $T \in \mathbb{N}_0$ trials do not reach the reward goal λ_0 .

Note that the decision maker can observe success or failure after every decision, so we denote $m_a \in \mathbb{N}_0$ and $n_a \in \mathbb{N}_0$ as the numbers of successes and failures of arm $a \in \{1, 2\}$, respectively, at the $(m_1 + n_1 + m_2 + n_2)$ -th epoch.

Thus, the bandit problem can be formulated as a POSMDP:

- (a) $E_X = I := \mathbb{N}_0^2 \times \mathbb{N}_0^2$ is the observable state space. $x = (m_1, n_1, m_2, n_2) \in I$ represents the numbers of successes and failures of arm $a \in \{1, 2\}$, respectively, at the $(m_1 + n_1 + m_2 + n_2)$ -th decision epoch;
- (b) $E_Y = \Theta := [0, 1]^2$ is the unobserved state space. For each $y = (\theta_1, \theta_2) \in \Theta$, the components θ_1, θ_2 denote the unknown success probabilities of arm 1 and 2, respectively;
- (c) $A \equiv A(x) = \{1, 2\}$ is the action space, where $a \in A$ denotes the index of arm chosen;
- (d) Q_0 is the initial distribution of $\theta = (\theta_1, \theta_2)$, which is assumed to be a product of two uniform distributions, i.e., $Q_0 = Be(1, 1) \otimes Be(1, 1)$;
- (e) $R_0(d\lambda) = \delta_{\lambda_0}(d\lambda)$ is the initial distribution of the profit goal;
- (f) $r(x, y, a) \in \{0, 1\}$ denotes the reward after each decision action which represents the decision maker gets zero unit reward respect to failure or gets one unit reward respect to success.

In addition, we introduce the following random variables:

- (g) $Z := \{0, 1\}$ represents the observation after each trial, corresponding to success ($z = 1$) and failure ($z = 0$), where the distribution of the observation $z \in \{0, 1\}$ is based on the unobserved state $y = (\theta_1, \theta_2)$ and the action a , i.e., $q^Z(1|y, a) := \theta_a$ and $q^Z(0|y, a) := 1 - \theta_a$.

With reference to the analysis in [1], we can similarly derive that the conditional distribution $\mu(\cdot|x)$ of $y = (\theta_1, \theta_2)$ is the product of two Beta distributions $Be(m_1 + 1, n_1 + 1) \otimes Be(m_2 + 1, n_2 + 1)$. So we can rewrite the distribution of the observation z based on the observable state x and the action a by $q^Z(1|x, a) = \frac{m_a + 1}{m_a + n_a + 2} =: p_a(x)$ and $q^Z(0|x, a) = \frac{n_a + 1}{m_a + n_a + 2} =: 1 - p_a(x)$.

Then, we can construct the following semi-Markov kernel:

- (h) $Q(t, x', y'|x, y, a) = \sum_{z \in Z} F(t|x, y, a) \mathbb{I}_{\{\Phi(x, a, z)\}}(x') q^Z(z|x, a) \mu(y'|x')$ is the semi-Markov kernel where $F(dt|x, y, a) := \mathbb{I}_{\{t_a\}}(dt)$ and $\Phi(x, a, z) := \mathbb{I}_{\{1\}}(z)(x + e_{2a-1}) + \mathbb{I}_{\{0\}}(z)(x + e_{2a})$, where e_a is a 4-dimensional row vector with the a -th element being 1 and the others being 0. Specifically, we can get $Q(t, (m'_1, n'_1, m'_2, n'_2), (\theta'_1, \theta'_2)|(m_1, n_1, m_2, n_2), (\theta_1, \theta_2), a) = \mathbb{I}_{\{t_a\}}(dt) \times \frac{\theta'_1 m'_1 (1 - \theta'_1)^{n'_1}}{B(m'_1 + 1, n'_1 + 1)} \times \frac{\theta'_2 m'_2 (1 - \theta'_2)^{n'_2}}{B(m'_2 + 1, n'_2 + 1)} \times (\frac{n_a + 1}{m_a + n_a + 2} \mathbb{I}_{\{x + e_{2a}\}}(x') + \frac{m_a + 1}{m_a + n_a + 2} \mathbb{I}_{\{x + e_{2a-1}\}}(x'))$.

Furthermore, we can take the transition equations of the remaining profit goal λ at each stage according to the observation z : $\lambda' = \Psi(\lambda, a, z) = \mathbb{I}_{\{1\}}(z) \cdot (\lambda - 1) + \mathbb{I}_{\{0\}}(z) \cdot \lambda$. Then we rewrite the distribution of the remaining profit goal λ based on the observable state x by using $q^Z(\cdot|x, a)$. Thus, in this model, we can know the distribution of the unobservable state y and the remaining profit goal λ and both can be rewritten as the conditional distribution of the observable state x . Thus, the above model can be formulated as an SMDP and we can rewrite the optimality equation as

$$\begin{aligned} V(t, x, \lambda) &= \min_{a \in A} \left\{ \sum_z q^Z(z|x, a) V((t - t_a)^+, \Phi(x, a, z), \Psi(\lambda, a, z)) \right\} \\ &= \min_{a \in A} \left\{ p_a(x) V((t - t_a)^+, x + e_{2a-1}, \lambda - 1) \right. \\ &\quad \left. + (1 - p_a(x)) V((t - t_a)^+, x + e_{2a}, \lambda) \right\} \\ &=: \min_{a \in A} T^a V(t, x, \lambda) \quad \forall (t, x, \lambda) \in \mathbb{N}_0 \times I \times \mathbb{N}_0, \end{aligned} \quad (5)$$

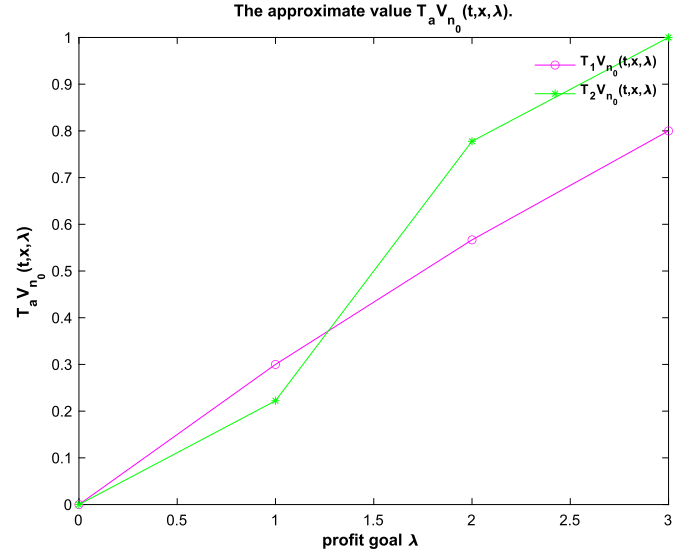


Fig. 1. The approximate value function $T_a V_{n_0}(t, x, \lambda)$ w.r.t. λ .

satisfying $V(t, x, \lambda) = 0$ with $\lambda < 0$ and $V(T, x, \lambda) = \mathbb{I}_{[0, +\infty)}(\lambda)$.

Based on Theorem 12, we can derive an approach of computing ϵ -optimal policies.

Algorithm 1.

Approach for computing ϵ -optimal policies:

Step 1 (Initialization):

For each $(t, x, \lambda) \in \mathbb{N}_0 \times I \times \mathbb{N}_0$, choose $V_{-1}(t, x, \lambda) = \mathbb{I}_{[0, +\infty)}(\lambda)$ as the initial value and set $n = -1$.

Step 2 (Evaluation):

• For the system under value $V_n(t, x, \lambda)$, calculate the function $T^a V_n(t, x, \lambda)$ by (5) for all $a \in A(x)$.

Step 3 (Improvement):

• Update the value functions V_n and the actions f_n^* as follows.

$$V_{n+1}(t, x, \lambda) = \min_{a \in A(x)} \{T^a V_n(t, x, \lambda)\}$$

$$f_{n+1}^*(t, x, \lambda) = \arg \min_{a \in A(x)} \{T^a V_n(t, x, \lambda)\}$$

Step 4 (Stopping Rule):

• Given a sufficiently small number $\rho > 0$ and $\beta > 0$ in Theorem 12, if $n = n_0 = \lceil \log_{\beta} \rho^{(1-\beta)} \rceil + 1$, then stop because $0 \leq V^* - V_{n_0} \leq \rho$. Moreover, $f_{n_0}^*(t, x, \lambda)$ is an $\epsilon := \rho(2 - \beta)$ -optimal policy.

Next, we use MATLAB to implement this algorithm and find an ϵ -optimal policy. For the convenience of numerical calculation, we take the following parameter setting: $t_1 = 1, t_2 = 2, T = 5, \lambda_0 = 3$. With the setting above, Assumptions 6 and 9 hold in this example with $\delta = 0.9$ and $\epsilon_0 = 0.99$. Taking $\epsilon = 10^{-4}$, we get $n_0 = 28$. Therefore, the value iteration algorithm proposed above is valid and an ϵ -optimal policy is ensured by Theorem 12.

Thus, we can compute $V_{n_0}(t, x, \lambda)$ and the ϵ -optimal policy for any $(t, x, \lambda) \in \{0, \dots, T\} \times I \times \{0, \dots, \lambda_0\}$. We just take $x = (0, 0, 1, 0)$ and $t = 5$ for example, where other cases are followed similarly. We get Fig. 1 and Table 1 for the approximate value $T_a V_{n_0}(5, (0, 0, 1, 0), \lambda)$ and the ϵ -optimal policy $f_{n_0}^*(t, x, \lambda)$, respectively.

Remark 13. For the bandit problem in the semi-Markov setting, if we take a special distribution $F(t|x, y, a) = \mathbb{I}_{\{t_a\}}(t)$, we can construct the Beta conjugate prior. However, if we consider other distributions, such as exponential distribution, i.e., $F(t|x, y, a) = 1 - e^{-\theta_a t}$, we do not necessarily find a suitable conjugate family. In addition, for other prior distributions like Gamma or Normal distribution, if the sojourn time distribution takes the special form $F(t|x, y, a) = \mathbb{I}_{\{t_a\}}(t)$ as described above, we will find suitable conjugate family, but if the sojourn time follows other general distributions, we do not necessarily find a suitable conjugate family. To be specific, for Gamma prior distributions, we will find

Table 1
The ϵ -optimal policy $f_{n_0}^*(t, x, \lambda)$.

t	x	$\lambda = 0$	$\lambda = 1$	$\lambda = 2$	$\lambda = 3$
5	(0, 0, 0, 2)	1 or 2	1	1	1 or 2
5	(0, 0, 1, 0)	1 or 2	2	1	1
5	(2, 0, 1, 0)	1 or 2	1	1 or 2	1 or 2
4	(0, 1, 0, 0)	1 or 2	1	1	1
4	(0, 1, 1, 0)	1 or 2	1	1 or 2	1 or 2
3	(0, 0, 2, 0)	2	1	1 or 2	1 or 2
3	(0, 1, 0, 0)	1	1	1	1
3	(0, 2, 0, 0)	2	1	1 or 2	1 or 2
2	(0, 0, 1, 0)	1	1 or 2	1 or 2	1 or 2
2	(1, 2, 0, 0)	1	1 or 2	1 or 2	1 or 2
1	(1, 1, 0, 0)	1	1	1	1
1	(0, 2, 1, 0)	1	1	1	1

a class of likelihood function which is Poisson-distributed such that the posterior distribution is again Gamma-distributed. Moreover, a class of prior Normal distributions is conjugate to a class of normally-distributed likelihood functions.

Specifically, if the sojourn time in the transition kernel is assumed to be one unit, then it can be simplified to a discrete-time POMDP and we can simply write the optimality equation as

$$V_n(x, \lambda) = \min_{a \in A} \left\{ \sum_{z \in Z} Q^Z(z|x, a) V_{n-1}(\Phi(x, a, z), \Psi(\lambda, a, z)) \right\} \\ =: \min_{a \in A} Q_a V_{n-1}(x, \lambda), \quad \forall (x, \lambda) \in I \times \mathbb{N}_0, \quad (6)$$

satisfying $V_n(x, \lambda) = 0$ ($\lambda < 0$), $V_0(x, \lambda) = \mathbb{I}_{[0, +\infty)}(\lambda)$. We introduce a symbol $x^\pm := \max\{0, \pm x\}$ and get the following results for the POMDPs.

Theorem 14. Let $d_n := Q_2 V_{n-1} - Q_1 V_{n-1}$, $n \geq 0$. Then it holds:

a) The policy $\pi^* = (f_N^*, \dots, f_1^*)$ is optimal, where

$$f_n^*(x, \lambda) = \begin{cases} 1, & d_n(x, \lambda) > 0; \\ 2, & d_n(x, \lambda) \leq 0, \end{cases}$$

is the minimizer of (6).

b) The sequence $\{d_n\}$ satisfies the recursion: $d_1(x, \lambda) = (p_1(x) - p_2(x))(\mathbb{I}_{[0, \infty)}(\lambda) - \mathbb{I}_{[0, \infty)}(\lambda - 1))$ and $d_{n+1} = Q_1 d_n^+ - Q_2 d_n^-$.

c) The 'stay-on-a-winner' property holds for the optimal policy π^* , i.e., $f_{n+1}^*(x, \lambda) = a \implies f_n^*(x + e_{2a-1}, \lambda - 1) = a$.

Proof. a) It follows directly from the definition of d_n .

b) Since $V_0(x, \lambda) = \mathbb{I}_{[0, +\infty)}(\lambda)$, we get

$$d_1(x, \lambda) = Q_2 V_0(x, \lambda) - Q_1 V_0(x, \lambda) \\ = p_2(x) V_0(x + e_3, \lambda - 1) + (1 - p_2(x)) V_0(x + e_4, \lambda) \\ - p_1(x) V_0(x + e_1, \lambda - 1) - (1 - p_1(x)) V_0(x + e_2, \lambda) \\ = (p_1(x) - p_2(x))(\mathbb{I}_{[0, \infty)}(\lambda) - \mathbb{I}_{[0, \infty)}(\lambda - 1)).$$

For $n \geq 1$ we obtain that

$$d_{n+1} = Q_2 V_n - Q_1 V_n \\ = Q_2(Q_1 V_{n-1} - d_n^-) - Q_1(Q_2 V_{n-1} - d_n^+) \\ = Q_2 Q_1 V_{n-1} - Q_2 d_n^- - Q_1 Q_2 V_{n-1} + Q_1 d_n^+ \\ = Q_1 d_n^+ - Q_2 d_n^-.$$

c) We show that $f_{n+1}^*(x, \lambda) = 1$ implies $f_n^*(x + e_1, \lambda - 1) = 1$ and the statement for $a = 2$ can be shown in the same way. Now we assume that $f_{n+1}^*(x, \lambda) = 1$ and $f_n^*(x + e_1, \lambda - 1) = 2$. The last equation implies $d_n(x + e_1, \lambda - 1) \leq 0$, which leads to $d_n(x + e_2, \lambda) \leq 0$ by induction on n with b). Then we obtain

$$d_{n+1}(x, \lambda) = Q_1 d_n^+(x, \lambda) - Q_2 d_n^-(x, \lambda) \\ = p_1(x) d_n^+(x + e_1, \lambda - 1) + (1 - p_1(x)) d_n^+(x + e_2, \lambda) \\ - p_2(x) d_n^-(x + e_3, \lambda - 1) - (1 - p_2(x)) d_n^-(x + e_4, \lambda) \\ = -p_2(x) d_n^-(x + e_3, \lambda - 1) - (1 - p_2(x)) d_n^-(x + e_4, \lambda) \leq 0.$$

This is a contradiction to the assumption $f_{n+1}^*(x, \lambda) = 1$ and the statement follows.

CRediT authorship contribution statement

Xin Wen: Writing – original draft, Formal analysis, Data curation.
Xianping Guo: Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.
Li Xia: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported in part by the National Key Research and Development Program of China (2022YFA1004600) and the National Natural Science Foundation of China (72342006, 72301304, 72371253, 62073346). The work described in this paper was also partially supported by InnoHK initiative, The Government of the HKSAR, and Laboratory for AI-Powered Financial Technologies.

Appendix A

Proof of Lemma 8. (a). To show that $V_n^\pi(\cdot, \cdot, \cdot) \in \mathcal{H}$, it suffices to show that $V_n^\pi(\cdot, \cdot, \cdot)$ is Borel-measurable on $\mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$. We prove this by induction. When $n = -1$, it is obviously true. Now we assume $V_n^\pi(\cdot, \cdot, \cdot)$ is Borel-measurable for $n \geq -1$ and $\pi \in \Pi_M$. Then by definition of V_n^π , $\forall \pi = \{f_0, f_1, \dots\} \in \Pi_M$, we have

$$V_{n+1}^\pi(t, x, \mu) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda]}(r(x, y, f_0)t) (1 - Q(t, E_X, E_Y | x, y, f_0)) \times \\ \mathbb{I}_{[0, +\infty)}(\lambda) \mu(dy, d\lambda) + \int_{E_Y} \int_{\mathbb{R}} \int_{E_X} \int_0^t \mathbb{P}^{(1)\pi}_{(t-u, x', y', \lambda - r(x, y, f_0)u)} \times \\ (\sum_{m=0}^n \int_{S_m \wedge (t-u)}^{S_{m+1} \wedge t} r(X(s), Y(s), A(s)) ds \leq \lambda - r(x, y, f_0)u) Q^X(du, dx' | x, \mu^Y, f_0) \\ \mathbb{I}_{[0, +\infty)}(\lambda - r(x, y, f_0)u) \frac{\mu(dy, d\lambda)}{\mu^Y(dy)}. \text{ On the other hand, when } t \geq u, \text{ it can be} \\ \text{derived that } V_n^{(1)\pi}(t - u, x', \Psi(t, x, \mu, f_0, u, t - u, x')) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}^{(1)\pi}_{(t-u, x', y', \lambda')} \times \\ (\sum_{m=0}^n \int_{S_m \wedge (t-u)}^{S_{m+1} \wedge (t-u)} r(X(s), Y(s), A(s)) ds \leq \lambda') \times \\ \frac{\int_{E_Y} \int_{\mathbb{R}} q(u, x', y' | x, y, f_0) \eta(dy') \delta_{\{\lambda - r(x, y, f_0)u\}}(d\lambda') \mathbb{I}_{[0, \infty)}(\lambda') \mu(dy, d\lambda)}{\int_{E_Y} q^X(u, x' | x, y, f_0) \mu^Y(dy)} = \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \infty)}(\lambda - \\ r(x, y, f_0)u) \mathbb{P}^{(1)\pi}_{(t-u, x', y', \lambda - r(x, y, f_0)u)} (\sum_{m=0}^n \int_{S_m \wedge (t-u)}^{S_{m+1} \wedge (t-u)} r(X(s), Y(s), A(s)) ds \leq \\ \lambda - r(x, y, f_0)u) \frac{\mu(dy, d\lambda)}{\mu^Y(dy)}. \text{ Thus, we have } V_{n+1}^\pi(t, x, \mu) =$$

$\int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda]}(r(x, y, f_0)t) (1 - Q^X(t, E_X | x, y, f_0)) \mathbb{I}_{[0, +\infty)}(\lambda) \mu(dy, d\lambda) + \int_{E_X} \int_0^t V_n^{(1)\pi}(t - u, x', \Psi(t, x, \mu, f_0, u, t - u, x')) Q^X(du, dx' | x, \mu^Y, f_0) = T^{f_0} V_n^{(1)\pi}(t, x, \mu)$. So $V_{n+1}^\pi(t, x, \mu)$ is Borel-measurable and thus $V_n^\pi(\cdot, \cdot, \cdot)$ is Borel-measurable for all $n \geq -1$. Since a limit of measurable functions is still measurable, we have $V^\pi = \lim_{n \rightarrow \infty} V_n^\pi \in \mathcal{H}$.

(b). From the proof of (a), we have $V_{n+1}^\pi = T^{f_0} V_n^{(1)\pi}$. Letting $n \rightarrow \infty$, by the dominated convergence theorem we derive $V^\pi = T^{f_0} V^{(1)\pi}$. The last statement is obviously true.

Proof of Theorem 10. Under Assumption 9, by the measurable selection theorem [3], for any fixed $G \in \mathcal{G}$, there exists an $g \in \mathbb{F}$ such that

$TG(t, x, \mu) = T^g G(t, x, \mu) = \min_{a \in A(x)} \{T^a G(t, x, \mu)\}$ for each $(t, x, \mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$. Thus we have $F_n^* \in \mathcal{H}$ and TF_n^* is well defined for every $n \geq -1$. Next we prove the fact that for every $(t, x, \mu) \in \mathbb{R}_+ \times E_X \times \mathcal{P}(E_Y \times \mathbb{R})$ and $\forall n \geq -1$,

$$F_n^*(t, x, \mu) \geq F_{n+1}^*(t, x, \mu).$$

When $n = -1$, $TF_{-1}^*(t, x, \mu) = \inf_{a \in A(x)} \{ \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda]}(r(x, y, a)t)(1 - Q^X(t, E_X | x, y, a))\mu(dy, d\lambda) + \int_{E_X \times \mathbb{R}} \int_0^t Q^X(du, dx' | x, \mu^Y, a)\mathbb{I}_{[0, \infty)}(\lambda') \times \frac{\int_{E_Y} \int_{\mathbb{R}} q(u, x', y' | x, y, a)\eta(d'y')\mathbb{I}_{[0, \infty)}(\lambda - r(x, y, a)u)\mu(dy, d\lambda)}{\int_{E_Y} q^X(u, x' | x, y, a)\mu^Y(dy)} \} =$
 $\inf_{a \in A(x)} \{ \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda]}(r(x, y, a)t)(1 - Q^X(t, E_X | x, y, a))\mu(dy, d\lambda) + \int_{E_Y} \int_0^t \int_{\mathbb{R}} Q^X(du, E_X | x, y, a)\mathbb{I}_{[0, \infty)}(\lambda - r(x, y, a)u)\mu(dy, d\lambda) \} \leq$
 $\int_{\mathbb{R}} \mathbb{I}_{[0, \infty)}(\lambda)\mu^\Lambda(d\lambda) = F_{-1}^*(t, x, \mu)$. Thus $F_0^*(t, x, \mu) = TF_{-1}^*(t, x, \mu) \leq F_{-1}^*(t, x, \mu)$. According to the monotonicity of the operator T , we can obtain that $F_{n+1}^*(t, x, \mu) \leq F_n^*(t, x, \mu)$. So the limit $\lim_{n \rightarrow \infty} F_n^*(t, x, \mu) := F^*(t, x, \mu)$ exists. Thus, since a limit of measurable functions is still measurable, we get $F^*(t, x, \mu) \in \mathcal{H}$.

It is clear that $F_{n+1}^*(t, x, \mu) = TF_n^*(t, x, \mu) \leq T^a F_n^*(t, x, \mu)$ for all $(t, x, \mu) \in \mathbb{R}_+ \times E$ and $a \in A(x)$. By the dominated convergence theorem, it can be derived that $F^* \leq T^a F^*$. Since $a \in A(x)$ is arbitrary, we have $F^* \leq TF^*$. On the other hand, since $F_{n+1}^* = TF_n^* \leq F_n^*$, letting $n \rightarrow \infty$, we get $\lim_n TF_n^* = T \lim_n F_n^* \leq \lim_n F_n^*$. So $TF^* \leq F^*$. Therefore, we obtain $F^* = TF^*$. Next we will prove that $F^*(t, x, \mu) = V^*(t, x, \mu)$.

By the measurable selection theorem, there exists an $f \in \mathbb{F}$ such that $F^*(t, x, \mu) = T^{f(t, x, \mu)} F^*(t, x, \mu)$ for any $(t, x, \mu) \in \mathbb{R}_+ \times E$. Then we define a sequence of probability measure $\{\mu_n\}_{n \geq 0}$ satisfying the filtered equation with $\mu_0 = \mu$ and the policy $\pi = (g_0, g_1, \dots)$ where $g_n(h_n) = f(t, x_n, \mu_n(\cdot | h_n))$ for all $n \geq 0$. Thus, we have $F^*(t, x, \mu) = T^{f(t, x, \mu)} F^*(t, x, \mu) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda]}(r(x, y, f(t, x, \mu, f, \theta_1, t - \theta_1, x_1))) \times Q^X(d\theta_1, dx_1 | x, \mu^Y, f)$. Now we claim that the following statement holds.

$$F^*(t, x, \mu) = \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] + \mathbb{E}_{(t, x, \mu)}^\pi [F^*(t - s_n, x_n, \mu_n)\mathbb{I}_{\{s_n \leq t\}}] \quad \forall n \geq 1. \quad (7)$$

When $t = 0$, it is clear that $F^*(0, x, \mu) = \int_{\mathbb{R}} \mathbb{I}_{[0, \infty)}(\lambda)\mu^\Lambda(d\lambda)$.

When $t > 0$ and $n = 1$, $\mathbb{E}_{(t, x, \mu)}^\pi [F^*(t - \theta_1)^+, x_1, \mu_1)\mathbb{I}_{\{\theta_1 > t\}}] = \int_{E_Y} \int_{\mathbb{R}} \int_{E_X} F^*([t - \theta_1]^+, x_1, \mu, f, \theta_1, [t - \theta_1]^+, x_1)) \times Q^X(d\theta_1, dx_1 | x, y, f)\mathbb{I}_{[0, \infty)}(\lambda - r(x, y, f)(t \wedge \theta_1))\mathbb{I}_{\{\theta_1 > t\}}\mu(dy, d\lambda) = \int_{E_Y} \int_{\mathbb{R}} \int_{E_X} Q^X(d\theta_1, E_X | x, y, f)\mathbb{I}_{[0, \infty)}(\lambda - r(x, y, f)(t \wedge \theta_1))\mathbb{I}_{\{\theta_1 > t\}}\mu(dy, d\lambda) = \int_{E_Y} \int_{\mathbb{R}} (1 - Q^X(t, E_X | x, y, f))\mathbb{I}_{[0, \lambda]}(r(x, y, f)t)\mu(dy, d\lambda)$, and $\mathbb{E}_{(t, x, \mu)}^\pi [F^*([t - \theta_1]^+, x_1, \mu_1)\mathbb{I}_{\{\theta_1 \leq t\}}] = \int_{E_Y} \int_{\mathbb{R}} \int_{E_X} \int_0^t F^*(t - \theta_1, x_1, \Psi(t, x, \mu, f, \theta_1, [t - \theta_1]^+, x_1))Q^X(d\theta_1, dx_1 | x, y, f)\mu(dy, d\lambda) = \int_{E_X} \int_0^t F^*(t - \theta_1, x_1, \Psi(t, x, \mu, f, \theta_1, [t - \theta_1]^+, x_1))Q^X(d\theta_1, dx_1 | x, \mu^Y, f)$. Thus, we know (7) holds for $n = 1$ and we assume it holds for $n \geq 1$. If $s_n < t$, we denote $t_n := t_{n-1} - \theta_n = t - s_n$. Thus, $F^*(t, x, \mu) = \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] + \int_{E_X} \int_0^t \int_{E_X} \int_0^{t-\theta_1} \dots \int_{E_X} \int_0^{t-\sum_{i=1}^{n-1} \theta_i} \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda_n]}(r(x_n, y_n, f)(t_n)(1 - Q^X(t_n, E_X | x_n, y_n, f))\mu_n(dy_n, d\lambda_n) + \int_{E_X} \int_0^{t_n} F^*(t_{n+1}, x_{n+1}, \mu_{n+1})Q^X(d\theta_{n+1}, dx_{n+1} | x_n, \mu_n^Y, f))\delta_{\{[t_n - \theta_{n+1}]^+\}}(t_{n+1}) \times \delta_{\{[t - \sum_{i=1}^n \theta_i]^+\}}(t_n) \delta_{\{\lambda - \sum_{i=0}^{n-1} r(x_i, y_i, f)\theta_{i+1}\}}(\lambda_n)Q^X(d\theta_n, dx_n | x_{n-1}, \mu_{n-1}^Y, f) \times Q^X(d\theta_{n-1}, dx_{n-1} | x_{n-2}, \mu_{n-2}^Y, f) \times \dots \times Q^X(d\theta_1, dx_1 | x, \mu^Y, f) = \mathbb{E}_{(t, x, \mu)}^\pi [F^*([0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] + \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_{n+1}, \mu_{n+1})\mathbb{I}_{\{s_n < t, \theta_{n+1} > t - s_n\}}] + \mathbb{E}_{(t, x, \mu)}^\pi [F^*(t - s_{n+1}, x_{n+1}, \mu_{n+1})\mathbb{I}_{\{s_{n+1} \leq t\}}] = \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_{n+1}, \mu_{n+1})\mathbb{I}_{\{s_{n+1} > t\}}] + \mathbb{E}_{(t, x, \mu)}^\pi [F^*(t - s_{n+1}, x_{n+1}, \mu_{n+1})\mathbb{I}_{\{s_{n+1} \leq t\}}]$. Therefore, for each $n \geq 1$, we have

$$F^*(t, x, \mu) = \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] + \mathbb{E}_{(t, x, \mu)}^\pi [F^*(t - s_n, x_n, \mu_n)\mathbb{I}_{\{s_n \leq t\}}] \geq \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}].$$

Let $n \rightarrow \infty$, by Assumptions 6 and 9 with the dominated convergence theorem, we have

$$F^*(t, x, \mu) \geq \lim_{n \rightarrow \infty} \mathbb{E}_{(t, x, \mu)}^\pi [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] = \lim_{n \rightarrow \infty} \int_{E_Y} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}_{(t, x, y, \lambda)}^\pi [\mathbb{I}_{[0, +\infty)}(\lambda - \int_0^t r(X(s), Y(s), A(s))ds)\mathbb{I}_{\{s_n > t\}}] \times \mathbb{I}_{[0, \infty)}(\lambda)\mu(dy, d\lambda) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}_{(t, x, y, \lambda)}^\pi (\int_0^t r(X(s), Y(s), A(s))ds \leq \lambda)\mathbb{I}_{[0, \infty)}(\lambda)\mu(dy, d\lambda) = V^*(t, x, \mu) \geq V^*(t, x, \mu).$$

Moreover, for any $\pi' = \{g_0, g_1, \dots\} \in \Pi$, using the similarly technique as above, we can obtain

$$F^*(t, x, \mu) = T^{f(t, x, \mu)} F^*(t, x, \mu) \leq T^{g_0(\theta_0, t_0, x_0)} F^*(t, x, \mu) = \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(0, x_1, \mu_1)\mathbb{I}_{\{s_1 > t\}}] + \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(t - s_1, x_1, \mu_1)\mathbb{I}_{\{s_1 \leq t\}}].$$

When $s_1 \leq t$, we get

$$F^*(t - s_1, x_1, \mu_1) = F^*(t_1, x_1, \mu_1) = T^{f(t_1, x_1, \mu_1)} F^*(t_1, x_1, \mu_1) \leq T^{g_1(\theta_0, t_0, x_0, a_0, \theta_1, t_1, x_1)} F^*(t_1, x_1, \mu_1) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{I}_{[0, \lambda_1]}(r(x_1, y_1, g_1)t_1)(1 - Q^X(t_1, E_X | x_1, y_1, g_1))\mu_1(dy_1, d\lambda_1) + \int_{E_X} \int_0^{t_1} F^*(t_1 - \theta_2, x_2, \Psi(t_1, x_1, \mu_1, g_1, \theta_2, [t_1 - \theta_2]^+, x_2)) \times Q^X(d\theta_2, dx_2 | x_1, \mu_1^Y, g_1).$$

Discussing similarly as above, we can obtain $F^*(t, x, \mu) \leq \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(0, x_2, \mu_2)\mathbb{I}_{\{s_2 > t\}}] + \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(t - s_2, x_2, \mu_2)\mathbb{I}_{\{s_2 \leq t\}}]$. Furthermore, iterating n times, it can be derived that $F^*(t, x, \mu) \leq \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] + \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(t - s_n, x_n, \mu_n)\mathbb{I}_{\{s_n \leq t\}}]$.

Let $n \rightarrow \infty$, by Assumption 6 and Assumption 9 with the dominated convergence theorem, we have

$$F^*(t, x, \mu) \leq \lim_{n \rightarrow \infty} \mathbb{E}_{(t, x, \mu)}^{\pi'} [F^*(0, x_n, \mu_n)\mathbb{I}_{\{s_n > t\}}] = \lim_{n \rightarrow \infty} \int_{E_Y} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}_{(t, x, y, \lambda)}^{\pi'} [\mathbb{I}_{[0, +\infty)}(\lambda - \int_0^t r(X(s), Y(s), A(s))ds)\mathbb{I}_{\{s_n > t\}}] \times \mathbb{I}_{[0, \infty)}(\lambda)\mu(dy, d\lambda) = \int_{E_Y} \int_{\mathbb{R}} \mathbb{P}_{(t, x, y, \lambda)}^{\pi'} (\int_0^t r(X(s), Y(s), A(s))ds \leq \lambda)\mathbb{I}_{[0, \infty)}(\lambda)\mu(dy, d\lambda) = V^{\pi'}(t, x, \mu).$$

Since the arbitrariness of the policy $\pi' \in \Pi$, it can be derived $F^*(t, x, \mu) \leq V^*(t, x, \mu)$. Thus, we have $F^*(t, x, \mu) = V^*(t, x, \mu)$. Furthermore, $V^*(t, x, \mu) \in \mathcal{H}$.

(b). According to part (a), we get $V^*(t, x, \mu) = TV^*(t, x, \mu)$. For any function $V(t, x, \mu)$ satisfying $V(t, x, \mu) = TV(t, x, \mu)$, replacing $F^*(t, x, \mu)$ to $V(t, x, \mu)$ in part (a), we can obtain $V(t, x, \mu) = V^*(t, x, \mu)$ as the proof of part (a). Thus, the uniqueness of the solution to the optimality equation has been proved.

(c). Using the similar proof as part (a), replacing $F^*(t, x, \mu)$ to $V^*(t, x, \mu)$ and $\pi = f$ to $\pi^* = f^*$ in part (a), we have $V^*(t, x, \mu) \geq V^{\pi^*}(t, x, \mu)$. On the other hand, $V^*(t, x, \mu) = \inf_{\pi \in \Pi} V^\pi(t, x, \mu) \leq V^{\pi^*}(t, x, \mu)$, so we have $V^*(t, x, \mu) = V^{\pi^*}(t, x, \mu)$ and (c) is proved.

References

- [1] N. Bäuerle, U. Rieder, *Markov Decision Processes with Applications to Finance*, Springer Science & Business Media, 2011.
- [2] N. Bäuerle, U. Rieder, Partially observable risk-sensitive Markov decision processes, *Math. Oper. Res.* 42 (4) (2017) 1180–1196.
- [3] D. Bertsekas, S. Shreve, *Stochastic Optimal Control: the Discrete-Time Case*, Athena Scientific, 1996.
- [4] M. Bouakiz, Y. Kebir, Target-level criterion in Markov decision processes, *J. Optim. Theory Appl.* 86 (1) (1995) 1–15.
- [5] R. Cavazos-Cadena, D. Hernández-Hernández, Successive approximations in partially observable controlled Markov chains with risk-sensitive average criterion, *Stoch. Int. J. Probab. Stoch. Process.* 77 (6) (2005) 537–568.
- [6] A. Deep, S. Zhou, D. Veeramani, Y. Chen, Partially observable Markov decision process-based optimal maintenance planning with time-dependent observations, *Eur. J. Oper. Res.* 311 (2) (2023) 533–544.
- [7] L. Di Masi GB Stettner, Risk-sensitive control of discrete-time Markov processes with infinite horizon, *SIAM J. Control Optim.* 38 (1) (1999) 61–78.
- [8] E.A. Feinberg, P.O. Kasyanov, M.Z. Zgurovsky, Partially observable total-cost Markov decision processes with weakly continuous transition probabilities, *Discrete Appl. Math.* 41 (2016) 656–681.
- [9] E.A. Feinberg, P.O. Kasyanov, M.Z. Zgurovsky, Markov decision processes with incomplete information and semiuniform Feller transition probabilities, *SIAM J. Control Optim.* 60 (4) (2022) 2488–2513.
- [10] D. Hernández-Hernández, Partially observed control problems with multiplicative cost, *Stoch. Int. J. Probab. Stoch. Process.* 41 (55) (1999).
- [11] O. Hernández-Lerma, *Adaptive Markov Control Processes*, Springer Science & Business Media, 2012.

- [12] K. Hinderer, Foundations of Non-stationary Dynamic Programming with Discrete Time Parameter, Springer, 1970, pp. 78–83.
- [13] X. Huang, X. Guo, X. Wen, Zero-sum games for finite-horizon semi-Markov processes under the probability criterion, *IEEE Trans. Autom. Control* 68 (9) (2023) 5560–5567.
- [14] Y. Huang, X. Guo, Finite horizon semi-Markov decision processes with application to maintenance systems, *Eur. J. Oper. Res.* 212 (1) (2011) 131–140.
- [15] Y. Huang, X. Guo, X. Song, Performance analysis for controlled semi-Markov systems with application to maintenance, *J. Optim. Theory Appl.* 150 (2) (2011) 395–415.
- [16] M.R. James, J.S. Baras, R.J. Elliott, Risk-sensitive control and dynamic games for partially observed discrete-time nonlinear systems, *IEEE Trans. Autom. Control* 39 (4) (1994) 780–792.
- [17] D. Rhenius, Incomplete information in Markovian decision models, *Ann. Stat.* 2 (6) (1974) 1327–1334.
- [18] L. Stettner, Risk sensitive portfolio optimization with completely and partially observed factors, *IEEE Trans. Autom. Control* 49 (3) (2004) 457–464.
- [19] X. Wen, X. Guo, L. Xia, Optimal dividend problems with a risk probability criterion, *Nav. Res. Logist.* 69 (2022) 421–430.
- [20] D.J. White, Minimizing a threshold probability in discounted Markov decision processes, *J. Math. Anal. Appl.* 173 (2) (1993) 634–646.
- [21] A. Yushkevich, Reduction of a controlled Markov model with incomplete data to a problem with complete information in the case of Borel state and control space, *Theory Probab. Appl.* 21 (1) (1976) 153–158.